ASSESSMENT 2 REPORT: Development of an LLM-Based Intelligent Agent for the Kaggle LLM 20 Questions Competition

MCTA 4363 DEEP LEARNING

SECTION 1

SEMESTER 2 25/26

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1. Introduction

Large Language Models (LLMs) have demonstrated significant capabilities in reasoning, question answering, and decision-making tasks. The Kaggle LLM 20 Questions Competition provides a challenging environment where an intelligent agent must identify a hidden keyword by asking strategic yes/no questions within a maximum of 20 rounds.

The objective of this project is to develop and improve an LLM-based Questioner Agent using the Gemma 7B Instruct model provided in the Kaggle starter notebook. The project focuses on iterative improvements through prompt engineering, few-shot learning, output sanitization, and rule-based reasoning enhancements. The performance of each version is analyzed and compared with the original baseline implementation.

2. Competition Overview

The Kaggle LLM 20 Questions Competition is a language-based reasoning challenge in which two agents interact:

Questioner Agent

- Generates yes/no questions.
- Makes keyword guesses.
- Attempts to identify the hidden keyword using the fewest possible rounds.

Answerer Agent

- Receives the hidden keyword.
- Responds only with "yes" or "no".
- Provides feedback based on the correctness of the question.

The ultimate goal is to correctly identify the hidden keyword before reaching the 20-question limit.

3. Dataset Description

Unlike traditional machine learning competitions, the Kaggle LLM 20 Questions Competition does not provide a conventional training and testing dataset.

Instead, the competition operates using a hidden keyword repository maintained within the Kaggle environment. During each game, a keyword is randomly selected from predefined categories such as:

- Person
- Place
- Thing

The Questioner Agent must infer the keyword through strategic questioning and logical reasoning.

Information available to the agent includes:

- Previous questions asked
- Previous answers received
- Current game state
- Turn type (ask, answer, or guess)

Therefore, the project focuses on improving agent reasoning and decision-making rather than traditional data preprocessing or feature engineering.

4. Baseline Model

The baseline implementation was based on the official Kaggle Gemma 7B starter notebook.

Baseline Characteristics

- Gemma 7B Instruction-Tuned Model
- Standard prompt structure
- Basic question generation
- Simple keyword guessing mechanism

Observed Weaknesses

- Random and inconsistent questions
- Poor utilization of previous context
- Repeated questions
- Formatting artifacts in outputs
- Weak keyword extraction performance

Example outputs showed that the model frequently generated vague or unrelated questions that contributed little to narrowing the search space.

5. Iterative Model Development

Version 1 (V1): Prompt Engineering

Objective

Improve the reasoning quality of generated questions.

Modifications

Additional instructions were added to the Questioner prompt:

- Use previous questions and answers as context.
- Avoid repeated questions.
- Focus on narrowing the search space.
- Generate only yes/no questions.
- Prioritize informative questions.

Results

The generated questions became more structured and context-aware compared with the baseline implementation.

Version 2 (V2): Few-Shot Learning

Objective

Provide examples of desirable behavior.

Modifications

Few-shot examples were introduced into the prompt:

Example sequence:

Question: Is it a place?
Answer: Yes

Question: Is it a country?
Answer: Yes

Question: Is it located in Africa?
Answer: Yes

Guess: Tanzania

Additional examples covering multiple categories were also introduced.

Results

The model demonstrated improved consistency and better adherence to expected output formats.

Version 3 (V3): Output Sanitization

Objective

Remove unwanted formatting artifacts.

Modifications

Regular-expression-based post-processing was implemented to remove phrases such as:

- "Question:"

- "Sure, here's your question:"
- "The answer is:"
- "Guess the keyword:"

Results

Outputs became cleaner and more competition-compliant.

Keyword extraction also became more reliable.

Version 4 (V4): Rule-Based Guardrails

Objective

Improve reasoning structure and reduce inefficient questioning.

Modifications

Additional reasoning constraints were introduced:

- Identify whether the keyword is a person, place, or thing first.
- Ask broad questions before specific questions.
- Avoid subjective questions.
- Avoid low-information questions.
- Use the previous context before generating a new question.

Fallback keyword extraction logic was also added when the model failed to follow formatting requirements.

Results

The model demonstrated more category-oriented reasoning and improved robustness.

Although repeated and irrelevant questions occasionally remained, the overall questioning strategy became more structured compared with previous versions.

6. Evaluation Methodology

Because the competition does not provide traditional evaluation datasets, qualitative evaluation was performed through gameplay simulations.

The following criteria were used:

Question Relevance

Measures whether generated questions contribute to identifying the keyword.

Reasoning Structure

Measures whether questions follow a logical narrowing process.

Output Formatting

Measures compliance with required output formats.

Keyword Guessing Consistency

Measures the reliability of generated guesses.

Context Utilization

Measures how effectively previous questions and answers are incorporated.

7. Experimental Results

Version	Main Improvement	Observation
Baseline	Original Gemma 7B	Random questioning
V1	Prompt Engineering	More structured questions
V2	Few-Shot Learning	Better reasoning examples

V3	Output Sanitization	Cleaner outputs
V4	Rule-Based Guardrails	More consistent reasoning

Overall, each iteration contributed to improving the behavior of the Questioner Agent.

The most significant improvements were observed in prompt engineering and rule-based reasoning constraints.

8. Discussion

The experiments demonstrate that LLM behavior can be significantly influenced without modifying model weights.

Prompt engineering proved effective in guiding the model toward more strategic questioning patterns.

Few-shot learning improved consistency by demonstrating desired behavior.

Output sanitization ensured cleaner responses and improved competition compliance.

Rule-based guardrails further enhanced reasoning structure by encouraging category-based narrowing strategies.

However, several limitations remain. The model occasionally generates irrelevant questions, repeats previous questions, and produces inconsistent keyword guesses.

9. Limitations

The current implementation has several limitations:

- Repeated questions may still occur.
- Irrelevant questions are occasionally generated.
- Keyword extraction remains imperfect.
- Performance is below the top leaderboard solutions.

- No quantitative leaderboard evaluation was available due to competition submission restrictions.

10. Future Work

Future improvements may include:

- Memory mechanisms to prevent repeated questions.
- Additional few-shot examples across more categories.
- Information-gain-based question generation.
- Hybrid LLM and algorithmic search approaches.
- Quantitative evaluation using leaderboard scores.

11. Conclusion

This project successfully developed and improved an LLM-based Questioner Agent for the Kaggle LLM 20 Questions Competition using the Gemma 7B model.

Through multiple iterations involving prompt engineering, few-shot learning, output sanitization, and rule-based reasoning enhancements, the agent demonstrated improved questioning structure, cleaner outputs, and more consistent reasoning behavior.

Although challenges remain, the project highlights the effectiveness of prompt-level optimization techniques in improving LLM agent performance without modifying model parameters.